MATH 128a: Homework #12

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Due by: **Wednesday, December 3, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Numerical Quadrature 1

Derive a formula for approximating

$$\int_1^3 f(x)dx,$$

in terms of $f(0)$, $f(1)$ and $f(4)$. It should be exact for all polynomials of degree up to 2.

Answer of Problem 1

We look for a quadrature rule of the form

$$\hat{I}_{[1,3]}(f) = w_0 f(0) + w_1 f(1) + w_2 f(4).$$

Since the quadrature rule should be exact for polynomials up to degree 2, it should in particular integrate $1, x, x^2$ exactly. Hence

$$\int_1^3 x^i dx = w_0\, 0^i + w_1 1^i + w_2 4^i$$

for $i = 0, 1, 2$ ($0^0 = 1$ for convention). This yields a linear system of equations for the w_i:

$$\begin{aligned} w_0 + w_1 + w_2 &= 1 \\ w_1 + 4w_2 &= 4 \\ w_1 + 16w_2 &= \frac{26}{3} \end{aligned}$$

Using Gaussian elimination or Matlab, we find $w_0 = -5/6$, $w_1 = 22/9$ and $w_2 = 7/18$.

Problem 2: Numerical Quadrature 2

Derive the Newton-Cotes formula for

$$\int_0^1 f(x)dx,$$

based on the Lagrange interpolation polynomial at the nodes -2, -1 and 0. Apply this result to evaluate the integral when $f(x) = \sin(x)$.

Answer of Problem 2

We are looking for a quadrature rule of the form:

$$\hat{I}_{[0,1]}(f) = w_0 f(-2) + w_1 f(-1) + w_2 f(0).$$

We have seen in class that the weights are given by

$$w_i = \int_0^1 \ell_i(x) dx,$$

where $\ell_i(x)$ is the ith Lagrange polynomial. Inserting the quadature points x_i and evaluating the coefficients, we get

$$\ell_0(x) = \frac{x(x+1)}{2}, \quad \ell_1(x) = -x(x+2), \quad \ell_2(x) = \frac{(x+2)(x+1)}{2}.$$

Computing the integrals of these polynomials, we obtain

$$w_0 = \frac{5}{12}, \quad w_1 = -\frac{4}{3}, \quad w_2 = \frac{23}{12}.$$

Problem 3: Numerical Quadrature 3

Find a quadrature rule

$$\widehat{I}_{[0,1]}(f) = w_0 f(0) + w_1 f(1),$$

that approximates the integral

$$I_{[0,1]}(f) = \int_0^1 f(x) dx,$$

and that is exact for all functions of the form $f(x) = ae^x + b\cos(\pi x/2)$; $a, b \in \mathbb{R}$.

Answer of Problem 3

Find weights w_0, w_1 such that

$$\int_0^1 f(x)\, dx \approx w_0 f(0) + w_1 f(1)$$

is exact for all

$$f(x) = ae^x + b\cos\left(\frac{\pi x}{2}\right).$$

Exactness for e^x requires

$$\int_0^1 e^x\, dx = e - 1 = w_0 + w_1 e.$$

Exactness for $\cos(\pi x/2)$ requires

$$\int_0^1 \cos\left(\frac{\pi x}{2}\right)\, dx = \frac{2}{\pi} = w_0.$$

Thus

$$w_0 = \frac{2}{\pi}, \qquad w_1 = \frac{e - 1 - w_0}{e} = \frac{e - 1 - \frac{2}{\pi}}{e}.$$

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The quadrature rule is therefore

$$\int_0^1 f(x)\,dx \approx \frac{2}{\pi}\,f(0) + \frac{e-1-\frac{2}{\pi}}{e}\,f(1).$$

Problem 4: Numerical Quadrature 4

a) Use the Lagrange interpolation polynomial to derive a quadrature rule of the form

$$\widehat{I}_{[0,1]}(f) = w_0 f(1/3) + w_1 f(2/3),$$

that approximates the integral

$$I_{[0,1]}(f) = \int_0^1 f(x)dx.$$

b) Transform the resulting formula to one for integration over $[a, b]$, $a < b$.

c) Apply this result to evaluate

$$\int_0^\pi \sin(x)dx.$$

Compare with the exact value of the integral.

Answer of Problem 4

(a). Interpolation nodes: $x_0 = \frac{1}{3}$, $x_1 = \frac{2}{3}$.
The Lagrange basis polynomials are

$$L_0(x) = 2 - 3x, \qquad L_1(x) = 3x - 1.$$

Therefore,

$$w_0 = \int_0^1 L_0(x)\,dx = \frac{1}{2}, \qquad w_1 = \int_0^1 L_1(x)\,dx = \frac{1}{2}.$$

Thus the rule is

$$\int_0^1 f(x)\,dx \approx \frac{1}{2}\left(f\left(\tfrac{1}{3}\right) + f\left(\tfrac{2}{3}\right)\right),$$

which is exact for all linear functions.
(b). Apply the linear change of variables $x = a + (b-a)t$. The nodes become

$$x_0 = a + \frac{b-a}{3}, \qquad x_1 = a + \frac{2(b-a)}{3},$$

and the weights scale by $(b-a)$.
Thus

$$\int_a^b f(x)\,dx \approx \frac{b-a}{2}\left[f\left(a + \tfrac{b-a}{3}\right) + f\left(a + \tfrac{2(b-a)}{3}\right)\right].$$

(c). Apply to $\int_0^\pi \sin x\,dx$.

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Nodes:

$$x_0 = \frac{\pi}{3}, \qquad x_1 = \frac{2\pi}{3}.$$

Thus

$$\int_0^\pi \sin x \, dx \approx \frac{\pi}{2}\left(\sin\frac{\pi}{3} + \sin\frac{2\pi}{3}\right) = \frac{\pi\sqrt{3}}{2} \approx 2.7207.$$

The exact value is 2, so this rule overestimates the integral.

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